Separating B_1^+ and T_1 in MPME imaging: an intuitive account

Companion to the MPME technical note

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Abstract

The multi-pathway multi-echo (MPME) sequence of Cheng et al. [1] estimates the transmit scale B_1^+ and T_1 from two steady-state scans. This note explains, without proofs, why separating the two is hard and what makes it possible. One scan sees its flip angle only relative to the tissue's Ernst angle, so a larger flip angle in a tissue with a shorter T_1 looks exactly like a smaller one in a tissue with a longer T_1, at any flip angle and whatever is assumed about B_1^+ across the image. Two scans separate them only to the extent that their samples move at different rates when B_1^+ changes, a difference we call the lever. The lever predicts the B_1^+ bound to within a factor of about five over four orders of magnitude, explains why small flip angles fail and the 330° pulse works, and also governs the aliases. The proofs and the full analysis are in the technical note [2]; the choice of flip angles is in the design note [3].

The answer in brief

1. **One scan cannot separate M_0, B_1^+ and T_1, at any flip angle** (Section 2). It measures only the offset of its flip angle from the Ernst angle and an amplitude. More echoes, a smooth or shared B_1^+, or more physics in the model do not change this in practice.

2. **Two scans separate them through the lever** $L = c(x_2) - c(x_1)$, $c(x) = x/\sin x$, the difference in how fast the two samples move with B_1^+ (Section 3). Two small flip angles have $L \approx 0$; a 330° pulse has $L = -12.5$.

3. **With the published protocol, B_1^+ is no longer the dominant limit on** T_1 (Section 4): knowing it exactly would lower the T_1 bound only by a factor of 1.45. A B_1^+ supplied from elsewhere is worth using with both scans only if it is more accurate than the B_1^+ the two scans measure themselves: their high-flip data amplify its error tenfold at $B_1^+ = 1$ and up to about 260-fold near the 360° crossing.

4. **Aliases are the other roots of one equation in** B_1^+ (Section 5). They lie beyond a value of B_1^+ at which a pulse crosses a multiple of 180°; the 360° crossing is where the best local B_1^+ measurement meets the worst global behaviour.

5. **Two systematic limits are often undetectable in the data** (Section 6): a pulse-ratio error, which every design passes to T_1 with a factor of at least 1.6, and model errors near 360°.

1 Setting

Each MPME scan records three configuration pathways, three echoes each, of an unbalanced steady-state gradient echo; the two scans share $T_R = 25\,\mathrm{ms}$ and the readout and differ in the nominal flip angle (15° and 330°); in the published protocol scan 2 covers only the centre of k-space and serves mainly to map B_1^+. The actual flip angle of scan i is $x_i = B_1^+ \alpha_i$. The unknowns are M_0, B_1^+, T_1, T_2, T_2^*, the off-resonance $\Delta\omega$ and a phase ϕ_0. Precision is measured by Cramér–Rao bounds on the logarithms of the positive parameters (relative standard deviations), per unit σ/M_0 for complex Gaussian noise; at $\mathrm{SNR}(M_0) = 300$ divide by 300. Numbers refer to the tissue used by Cheng et al. to optimise their protocol ($T_1/T_2/T_2^* = 1500/70/60\,\mathrm{ms}$) at $B_1^+ = 1$ unless stated. Everything assumes the ideal model of [2, Sec. 2] (instantaneous pulses, one flip angle per voxel); Section 6 says what happens outside it.

2 One scan sees only the offset from the Ernst angle

The Ernst angle $\alpha_E = \arccos \mathrm{e}^{-T_R/T_1}$ maximises the signal of a spoiled gradient echo; for $T_1 = 1500\,\mathrm{ms}$ it is 10.4°, for 850 ms 13.8°. Put flip angles on the log half-angle axis $s = \ln|\tan(x/2)|$. On this axis the steady state of one scan has a remarkably simple form [2, Thm. 1, Prop. 2, eq. 13]: every pathway amplitude is the tissue's *amplitude* $M_0 \tan(\alpha_E/2)$ times a function of one number (for given T_2 and off-resonance), the *offset* of the sample from the Ernst angle,

$$w = \ln|\tan(x/2)| - \ln\tan(\alpha_E/2).$$

The spoiled-signal curve is a bell-shaped bump ($\operatorname{sech} w$) of fixed shape centred on the Ernst angle (Fig. 1); the ratios between pathways tell where on the bump the scan samples, and the overall scale tells the bump's height. That is all a scan reveals about (M_0, B_1^+, T_1); the echo decays and phases are spent on T_2, T_2^*, $\Delta\omega$ and ϕ_0.

Why one scan cannot separate B_1^+ from T_1. Raise B_1^+: the sample moves to the right. Shorten T_1 by the right amount: the Ernst angle grows and the bump moves to the right by the same distance. Lower M_0: the height is restored. The whole picture translates rigidly and the data do not change, at any flip angle, exactly. For small angles and $T_R \ll T_1$, $B_1^+ \to \kappa B_1^+$ is compensated by $T_1 \to T_1/\kappa^2$ and $M_0 \to M_0/\kappa$, the familiar B_1^+ sensitivity of variable-flip-angle T_1 mapping [4] (at 15° the exact logarithmic slopes are -2.02 and -1.01 [2, eq. 17]). A 10% error in B_1^+ becomes an error of about -18% in T_1.

Nothing inside one scan rescues it. More echoes or pathways sample the same point of the same bump. A smooth (e.g. polynomial) B_1^+ field does not help either: every voxel has its own T_1 and M_0 to absorb the change, so N voxels give $2N$ equations for $2N + p$ unknowns, and any field fits the data exactly [2, Prop. 7, Cor. 3]. Voxels that share one B_1^+ slide together, keeping their M_0 ratios exactly and their T_1 ratios to about 10^{-4}, but their common scale stays free [2, Prop. 8]. Modelling more physics does not help in practice: relaxation during the pulse leaves the degeneracy exact, and a known slice profile, off-resonance during the pulse or diffusion give bounds on $\ln B_1^+$ 400 to 5×10^6 times larger than a second scan does, and no information at all as soon as the extra physics has an unknown parameter of its own [2, Sec. 5.4].

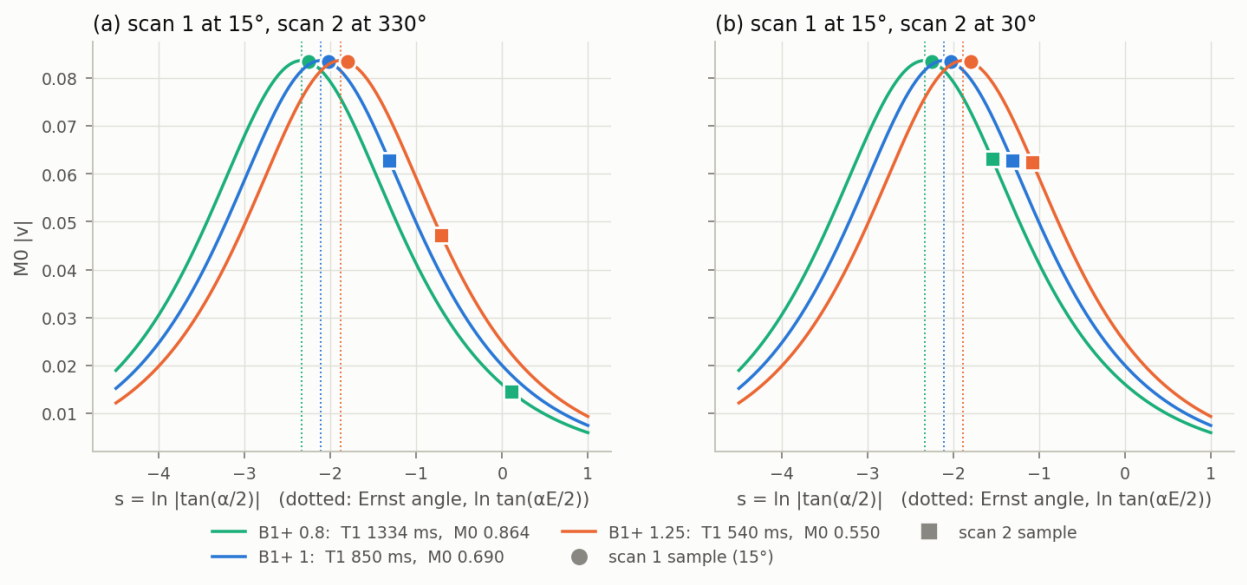


Figure 1: The signal amplitude on the log half-angle axis for three combinations of B_1^+, T_1 and M_0 that give identical scan-1 data (white matter, $T_1 = 850\,\text{ms}$; scan 1 at 15°). Raising B_1^+ moves the sample (circles) to the right; shortening T_1 moves the bump (dotted: Ernst angles) by the same amount; lowering M_0 keeps the height. Squares: the second scan, at 330° (a) or 30° (b). From [2, Fig. 2].

3 Two scans: the lever

A second scan samples the same bump at another point. When B_1^+ changes, each sample moves along the s-axis at its own *rate*

$$c(x) = \frac{\mathrm{d}\ln|\tan(x/2)|}{\mathrm{d}\ln B_1^+} = \frac{x}{\sin x} \qquad (x \text{ in radians}),$$

written $c_i = c(x_i)$ for scan i, while the bump (the tissue) moves the same for both. The compensation that worked for one scan now fails unless the two samples move at the same rate. B_1^+ is therefore visible only through the difference of the rates, the *lever* $L = c(x_2) - c(x_1)$: the bound on $\ln B_1^+$ is the uncertainty of the difference of the two offsets divided by $|L|$ [3, Prop. 3].

Figure 2b shows that this one number organises the problem: over 19 protocols the B_1^+ bound ranges from 1.1 to 13 000, while $|L|$ times the bound stays within a factor of about five.

Why small angles fail. For small angles $c(x) \approx 1 + x^2/6$, so two small flip angles have nearly the same rate and $L \approx (x_2^2 - x_1^2)/6 \to 0$. For 15°/30°, $L = 0.036$ and the B_1^+ bound is 1023: the parameters are identifiable in principle, but separating them needs an SNR of order 10^4. This is the leading-order scaling invariance $(M_0, B_1^+, T_1) \mapsto (M_0/\kappa, \kappa B_1^+, T_1/\kappa^2)$ of small-angle protocols [2, Prop. 4], seen as the limit $L \to 0$; adding more small angles does not help.

Why the 330° pulse works. A 330° pulse carries exactly the information of a 30° pulse about its own offset, but its rate is -11.5, so $L = -12.5$. The B_1^+ bound falls from 1023 to 2.9, by the factor $|L_{330}/L_{30}| = 351$ exactly [2, Prop. 5, Sec. 4.4]. Any second pulse whose rate differs strongly from the first scan's, and whose sample still sits on the informative part of the bump, would do

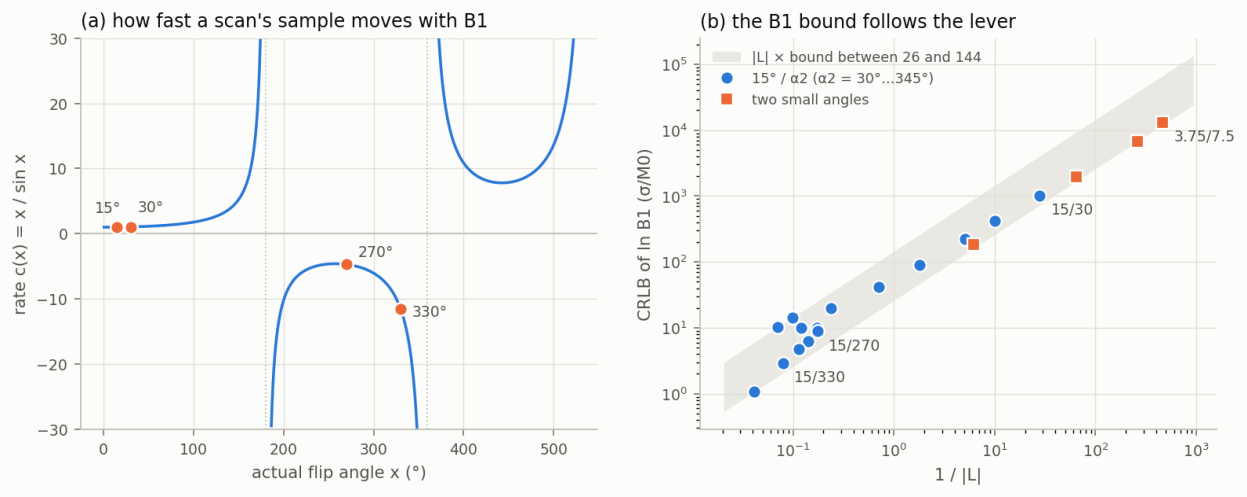


Figure 2: (a) The rate $c(x) = x/\sin x$ (dotted: multiples of 180°). Small angles move at rate ≈ 1; a 330° pulse is a 30° pulse in disguise ($|\tan 165°| = \tan 15°$) that moves eleven times faster and in the opposite direction. (b) Bound on $\ln B_1^+$ against $1/|L|$ for scan 1 at 15° with scan 2 from 30° to 345°, and for four pairs of small angles (all seven parameters free). The bound spans four orders of magnitude while $|L|$ times the bound stays between 26 and 144 (shaded); pulses with the same $|\tan(x/2)|$ (e.g. 30° and 330°) give identical products.

(where the samples sit accounts for the remaining factor of five in Fig. 2b); it does not need to be near 360° (a 270° pulse has rate −4.7).

An equation in B_1^+ alone. Each scan fixes its offset, i.e. the ratio $\xi_i = \tan(x_i/2)/\tan(\alpha_E/2)$ (up to sign). Dividing the two ratios eliminates T_1 and M_0:

$$f(B_1^+) = \frac{\tan(B_1^+\alpha_2/2)}{\tan(B_1^+\alpha_1/2)} = \frac{\xi_2}{\xi_1}, \text{ a number the data fix,} \tag{1}$$

the half-angle analogue of double-angle B_1^+ mapping. Its logarithmic slope is the lever itself,

$$\frac{\mathrm{d}\ln|f|}{\mathrm{d}\ln B_1^+} = L,$$

so the same number controls how precisely and how uniquely B_1^+ is determined (Section 5).

4 What limits T_1 once B_1^+ is resolved

Table 1 frees the parameters one group at a time (the factors depend on the order). With the published protocol, knowing B_1^+ exactly would lower the T_1 bound only by a factor of 1.45, against 75 for 15°/30°: most of the uncertainty now comes from the weak T_1 encoding of a 15° scan with $T_R \ll T_1$, shared with M_0, T_2 and T_2^*. The cost of not knowing B_1^+ has an exact and simple form,

$$\mathrm{Var}(\ln T_1) = \mathrm{Var}(\ln T_1 \mid B_1^+ \text{ known}) + \beta^2\,\mathrm{Var}(\ln B_1^+), \tag{2}$$

where β is the factor with which an error in a supplied B_1^+ reaches T_1 (next paragraph): for the published protocol $40.5^2 = 28.0^2 + (10.1 \times 2.9)^2$, and for 15°/30°, $\beta = -2.06$ but the B_1^+ bound is 1023.

Table 1: Bound on $\ln T_1$ (per unit σ/M_0) as parameters are freed in turn ($B_1^+ = 1$, reference tissue; the factors between rows depend on the order).

Unknown parameters	15°/330°	15°/30°
T_1 only	7.9	7.9
+ T_2, T_2^*, $\Delta\omega$, ϕ_0	21.1	21.1
+ M_0	28.0	28.0
+ B_1^+ (all free)	40.5	2106.1
scan 1 only, B_1^+ known	57.1	

A B_1^+ supplied from elsewhere. Suppose B_1^+ comes from a separate map with a small error and the rest is fitted. With scan 1 alone, the T_1 error is $\beta = -2.0$ times the B_1^+ error at every B_1^+, as in Section 2. With both scans the fit must reconcile two samples that move at different rates, and $\beta = -\bar{c}/h$, where $\bar{c}$ is the average of the two rates weighted by the information each scan carries about its offset and $h = \tau/(2\sinh\tau) \approx \frac{1}{2}$, $\tau = T_R/T_1$, is how far the bump moves per unit $\ln T_1$ (checked against direct computation to 10^{-15}). For the published protocol $\beta = +2.4$ at $B_1^+ = 0.8$ and $+10$ at $B_1^+ = 1$; it swings from about $+260$ to -270 within a fraction of a percent of the 360° crossing at $B_1^+ = 1.09$, and is -15 at $B_1^+ = 1.2$ (Fig. 3c). By (2), a supplied B_1^+ improves on the joint fit only if its error is below the joint B_1^+ bound ($2.9\,\sigma/M_0$, about 1% at SNR 300). Otherwise B_1^+ should be fitted jointly from both scans (40.5), or, when scan 2 is low-resolution as in the published protocol, the map should be used with scan 1 alone (57.1, error transfer -2).

5 Aliases and the 360° crossing

An *alias* is a different set of parameters that produces exactly the same noise-free data; no SNR separates it from the truth, only prior knowledge or more data. In MPME the aliases are simply the other roots of (1): any B_1^+ with the same ratio reproduces both scans once T_1 and M_0 are adjusted, provided the adjusted T_1 exists (its Ernst angle stays below 90°) [2, eq. 18]. A *complex alias* has the same signed ratio; a *magnitude alias* has the opposite sign, which negates scan 2 relative to scan 1 and is invisible in magnitude images. The alias values of B_1^+ depend on the true B_1^+ alone, not on the tissue.

For the published protocol at $B_1^+ = 1$ (Fig. 3a) the magnitude alias is $B_1^+ = 1.20$, inside the physiological range: the scan-2 pulse becomes $396° \equiv +36°$ instead of $330° \equiv -30°$. Complex data with a phase shared by the two scans remove it, and so, for magnitude data, does the sign of the relative phase of the two FID images (Cheng et al.'s FID-sign test). The complex alias is $B_1^+ = 2.01$, excluded by the fitting range $B_1^+ \le 540/330$ (scan-2 angle at most 540°). Because $\mathrm{d}\ln|f|/\mathrm{d}\ln B_1^+ = L$, aliases arise where f has zeros or poles, i.e. where a pulse crosses a multiple of 180°; a design whose two pulses stay within $(0°, 180°)$ and $(180°, 360°)$ over the fitting range has none at all [3, Prop. 5]. A 270° second pulse fitted within $B_1^+ \in [0.667, 1.333]$ is such a design.

The 360° crossing. The published second pulse reaches 360° at $B_1^+ = 1.09$. There the scan-2 signal vanishes, and scan 2 measures B_1^+ alone, through the derivative of a null: locally B_1^+ is determined best (0.09 against 2.9 at $B_1^+ = 1$, Fig. 3b). Everything else goes wrong there. The T_1 bound rises to the scan-1 value with B_1^+ known (56 against 41 at $B_1^+ = 1$), because scan 2 now informs B_1^+ alone and no longer helps with T_1, M_0, T_2 or T_2^*; the supplied-B_1^+ factor swings between about ± 260 (Fig. 3c); the magnitude alias comes within a few percent of the truth; the likelihood has distant near-equal maxima at the other scan-2 nulls, which account for the large errors that

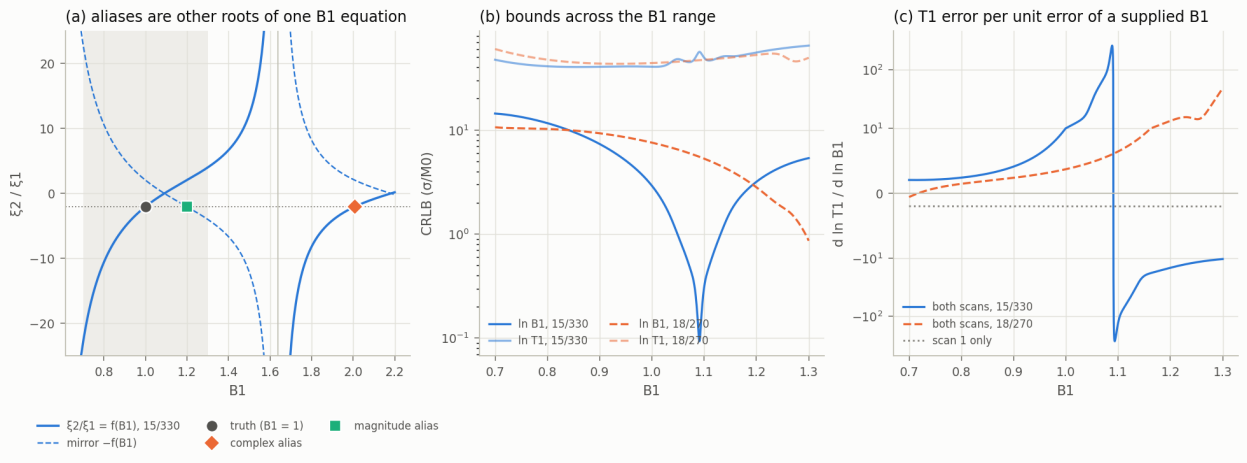


Figure 3: (a) The quantity (1) and its mirror image for 15°/330°; the horizontal line is the value at $B_1^+ = 1$, its other crossings are the aliases (grey band: $B_1^+ \in [0.7, 1.3]$; vertical line: 540/330). (b) Bounds on $\ln B_1^+$ and $\ln T_1$ across B_1^+ for 15°/330° (solid) and 18°/270° (dashed). (c) T_1 error per unit error of a supplied B_1^+, using both scans or scan 1 alone.

remain in the phantom after a spatial restart [2, Secs. 7 and 9]; and model errors bias T_1 most there (Section 6). A second pulse near 270° keeps the crossing outside $B_1^+ \in [0.7, 1.3]$ and has smoother bounds across the range (dashed), although its own crossing, at $B_1^+ = 1.33$, is close enough that its effects reappear near $B_1^+ = 1.3$.

6 Limits the bound cannot see

Pulse-ratio errors. If the two pulses do not have their nominal ratio, as with two pulses of different shape or duration, the fit absorbs the error exactly: two scans give three numbers for the three unknowns (M_0, B_1^+, T_1), so a shifted offset is reproduced by other values with zero residual. The T_1 error is at least 1.64 times the ratio error for every choice of angles (1.6–2.1 at $B_1^+ = 1$ for the shortlisted designs of [3, Table 2]; much more for small-angle pairs) [3, Prop. 4]. The ratio must be calibrated; the in-vivo factor $\beta = 1.24$ of [1], which multiplies this ratio, cannot be validated from the data.

Model errors near 360°. Near 360° the effective angle of the second pulse is small, so small imperfections change it by a large fraction. Unmodelled finite pulses (1 ms for 330° at a fixed peak B_1^+) bias T_1 by 10–20% where the scan-2 angle passes 330–350°, and a 10% flip-angle ramp across a voxel at the slab edge by 13–74% near or across 360°, often without a misfit that a χ^2 test would flag [3, Sec. 6]. Partial volume with CSF gives biased estimates with falsely confident uncertainty maps [2, Sec. 10].

7 Summary

Table 2 collects the limits and what does and does not remove them.

Estimators and design. The bounds are attainable: joint maximum likelihood on complex data reaches them (SD/bound 0.99–1.03 at SNR 1000–3000), and so does magnitude least squares with

Table 2: Limits of B_1^+/T_1 separation in MPME.

Limit	Origin	Removed by	Not removed by
One scan: exact degeneracy	data depend on the offset from the Ernst angle only	a second scan with a different rate	more echoes, smooth or shared B_1^+, more physics in the model
Small angles	equal rates, $L \approx 0$	a second pulse with a very different rate	more small-angle scans
Magnitude alias	sign of scan 2 lost	shared phase or FID sign; crossing-free fitting range	SNR
Complex alias	another root of (1)	a fitting range for B_1^+	SNR, shared B_1^+
Residual T_1 bound	confounding with T_2, T_2^*, M_0	SNR, priors	knowing B_1^+ (gains only 1.45)
Supplied B_1^+ error	high-flip data amplify it	fitting B_1^+ jointly; reduced to $\times 2$ with scan 1 alone	using it with both scans
Pulse-ratio error	absorbed exactly	calibration	any choice of angles, residuals
Model error near 360°	small effective angle	modelling the pulse; crossing outside the range	residuals (often)

the FID sign, while the published analytic inverse is 3–7 times above them, an algorithmic rather than an informational loss [2, Secs. 6–7]. For the choice of flip angles, the design note [3] finds the optimum flat around the published protocol: the angles buy robustness, not precision, and a second pulse near 255–270° with $\alpha_1 \approx 17$–$21°$, finite pulses in the model and a calibrated pulse ratio is the recommended default.

Reproducibility. Figures 2 and 3, Table 1 and the numbers of Sections 3 to 5 are generated by `scripts/limits_figures.py` (`make limits`); Fig. 1 by `scripts/single_scan_family.py`; (2) and the supplied-B_1^+ factor are unit tests in `tests/test_design.py`. The review-stage version of this note is preserved as the git tag `limits-review-record`.

References

[1] C.-C. Cheng, F. Preiswerk, W. S. Hoge, T.-H. Kuo, B. Madore. Multipathway multi-echo (MPME) imaging: all main MR parameters mapped based on a single 3D scan. *Magn. Reson. Med.* 81(3):1699–1713, 2019.

[2] S. Joshi. Multi-pathway multi-echo (MPME) relaxometry: forward model, information limits, and efficient estimation. Technical note, `docs/technote`.

[3] S. Joshi. Choosing the two flip angles of an MPME protocol: an optimal-design study. Technical note, `docs/flipnote`.

[4] S. C. L. Deoni, B. K. Rutt, T. M. Peters. Rapid combined T_1 and T_2 mapping using gradient recalled acquisition in the steady state. *Magn. Reson. Med.* 49(3):515–526, 2003.